

ANIMALS

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MORE THAN a million and a half species of animal have been identified, and

there are many millions more yet to be discovered. Animals are living organisms found in nearly all of the Earth's habitats, including the depths of the oceans, the freezing Arctic, and even inside other animals and plants. The animal kingdom is divided into animals without backbones (invertebrates), such as snails and lobsters, and animals with backbones (vertebrates), such as frogs and monkeys. Invertebrates make up 97 per cent of all animal species.



Black leopard

The leopard is a mammal. Its well-defined head is equipped with sense organs, including eyes, nose, tongue, and whiskers. Sharp teeth in the mouth allow the leopard to kill prey and tear off flesh. Muscular legs enable it to walk, run, and pounce.

What is an animal?

Animals are made up of many cells. Most move actively, and those that are fixed in one place, or sedentary, move their body parts. Animals live by taking food into their bodies. They have sensors and nervous systems that enable them to detect what is happening around them and respond appropriately.

Giant land flatworm



Animal classification

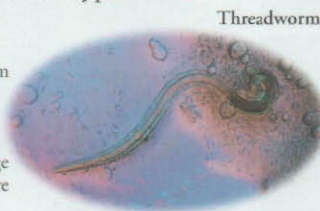
Animals are classified into groups according to their similarities and whether they have common ancestors. There are 35 major groups called phyla (singular phylum). Each phylum is divided into sub-groups. The smallest of these is the species, which contains animals of just one type.

Flatworms

These worms (phylum Platyhelminthes) have a flattened body with one opening, the mouth, on the underside. There are about 18,500 species including those, such as tapeworms, that are parasites of humans and other animals.

Nematodes

Roundworms, or nematodes (phylum Nematoda), have a thin, cylindrical body that is pointed at both ends. Free-living nematodes are found in many habitats and occur in very large numbers in soil. Many nematodes are parasites of plants and animals.



Threadworm

Sponge processed for human use



Sea anemones



Sponges

The simplest animals are sponges (phylum Porifera). There are about 5,000 species, most of which live in the sea attached to rocks and other objects. Water is drawn in through holes, or pores, in the sponge's body wall, and bits of food are filtered out and eaten by the sponge's cells.

Cnidarians

There are more than 9,000 species of cnidarians (phylum Cnidaria), most of which are found in the sea. They include jellyfish, sea anemones, hydras, and corals. Cnidarians catch food using tentacles armed with stinging threads, called nematocysts.

Annelids

Animals in the phylum Annelida include earthworms, marine bristleworms such as ragworms, and leeches. There are about 12,000 species, each of which has a body made up of segments with a mouth at one end and an anus at the other.



King ragworm

Echinoderms

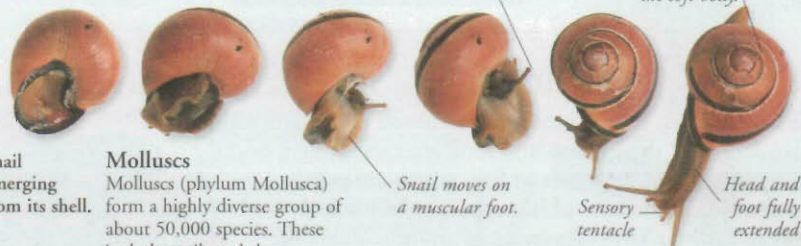
All echinoderms (phylum Echinodermata) live in the sea. The 6,500 or so species include sea urchins and starfish. Most have five parts radiating from a central point, hard plates under the skin, and many tube-feet.



Cushion star

Cushion star

Snail emerging from its shell.



Molluscs

Molluscs (phylum Mollusca) form a highly diverse group of about 50,000 species. These include snails and slugs, mussels and clams, and squids and octopuses. They are soft-bodied animals that may be protected by a shell. Most live in water, but some, such as snails, are found on land.

Stalked eye

Snail moves on a muscular foot.

Coiled shell protects the soft body.

Sensory tentacle

Head and foot fully extended

Arthropods

With at least one million known species, Arthropods (phylum Arthropoda) are the largest group of animals. They include insects, crustaceans (such as crabs), arachnids (such as spiders), and centipedes.

Arthropods have hard, jointed external skeletons.



Tarantula

Sharp teeth to grasp food

Caiman



Tail used for movement or balance is typical of many vertebrates.